

Revised Generation Interconnection System Impact Study Report

For Queue Project Q24-2381

State Street 100 kV
June 25, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2381. The Interconnection Customer proposes a 125.0 MW solar photovoltaic facility interconnecting at the State Street 100 kV substation in the Indiana area. The request was received on February 07, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 442 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,978,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
State Street / W Olympia 100 kV line	State Street / W Olympia 100 kV	126.9	216	5.8
State Street Substation / W Olympia 100 kV Ckt 2	West Olympia / State St 100 kV Ckt 1	126.9	216	5.8
State St Substation / W Olympia 100 kV #3	State St / West Olympia 100 kV	126.9	216	5.8
Monte Del Estado 6303 / Harper 100 kV	Dekalb / Richland 500 kV	105.6	532	6.1
Monroeville / Woodburn 100 kV Line	Richland / Dekalb 500 kV	103.6	398	5.1
Woodurn / Sage 100 kV L	Richland / Dekalb 500 kV	103.1	413	5.1
Monte Del Estado 6303 / Sage 100 kV L	Richland Substation / Dekalb 500 kV L	102.4	434	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
State Street / W Olympia Substation 100 kV	Rebuild line with higher-capacity conductor	\$1,049,000	33% (\$348,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
State Street Substation / W Olympia 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,194,000	33% (\$727,000)
W Olympia / State St 100 kV Line 3	Rebuild line with higher-capacity conductor	\$1,636,000	33% (\$542,000)
Harper / Monte DI Estado 6303 Substation 100 kV	Reconductor line	\$2,464,000	8% (\$186,000)
Monroeville / Woodburn 100 kV Line	Reconductor line	\$3,268,000	22% (\$727,000)
Sage / Woodburn 100 kV	Reconductor line	\$616,000	22% (\$137,000)
Sage / Monte Del Estado 6303 100 kV	Reconductor line	\$1,398,000	22% (\$311,000)
	Total allocated to project		\$2,978,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.